Worksheet 6.1

Science Quest 9: pages 240–1

Chemical building blocks

Student: ..... Class: .....

A history of ideas about the atom

The flow chart below shows how our knowledge about the atom developed. Draw your own diagrams to match the description of the changing model of matter.

About 400 BC: Greek philosopher Democritus first suggested that all substances consisted of tiny indestructible particles called atoms.

1808: John Dalton's atomic theory proposed that:
- all matter consisted of tiny particles called atoms
- atoms could not be divided into smaller particles
- atoms of the same element were alike
- atoms combined in simple whole number ratios.

1897: English scientist Sir J. J. Thomson explained that the atom contained negatively charged particles called electrons. His model suggested that atoms were positively charged spheres with negatively charged electrons embedded in them like the fruit in a plum pudding.

1911: Lord Rutherford proposed that the atom consisted mostly of empty space with a dense nucleus containing positively charged protons in the centre. Negatively charged electrons orbited the nucleus. Although Lord Rutherford's model of the atom was essentially the same as today's accepted model, Its one flaw was that it proposed that the orbiting electrons would eventually lose energy and spiral in towards the nucleus.

1913: Niels Bohr, a scientist who had studied with Rutherford, modified the model by suggesting that electrons orbit the nucleus at different energy levels. Only electrons with specific amounts of energy could exist at each level. His model proposed that electrons could move from one level to another by gaining or losing 'packets' of energy. Although Bohr's model explained why electrons did not spiral in towards the nucleus, it did not explain all of the known properties of atoms.

1932: Sir James Chadwick discovered that the nucleus contained particles called neutrons, as well as positively charged protons. Neutrons had no electric charge and a mass about the same as a proton.

© John Wiley & Sons Australia, Ltd